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Electric vehicles incorporating flatbeds and methods associated therewith

Abstract

In certain embodiments, an electric vehicle includes a front cage, a rear floor, an intermediate section, a utility cabinet, and a flatbed. In other embodiments, an electric vehicle includes a front cage, a rear floor, an intermediate section, and a flatbed. In some embodiments, the front cage at least partially defines an operator cabin, the rear floor is positioned rearward of the front cage in a longitudinal direction, and the intermediate section is disposed at least partially between the front cage and the rear floor in the longitudinal direction.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4705716	12/1986	Tang	N/A	N/A
4842326	12/1988	DiVito	296/10	B60P 3/42
4917430	12/1989	Lawrence	N/A	N/A
5082037	12/1991	Hammons et al.	N/A	N/A
5131714	12/1991	Evans, Sr. et al.	N/A	N/A
5363939	12/1993	Catlin	N/A	N/A
5642971	12/1996	Ragsdale	N/A	N/A
5988473	12/1998	Hagan et al.	N/A	N/A
5997227	12/1998	Bundy	N/A	N/A
6296301	12/2000	Schroeder et al.	N/A	N/A
D460039	12/2001	Smith	N/A	N/A
6464277	12/2001	Wilding	N/A	N/A
6926351	12/2004	Telehowski et al.	N/A	N/A
6935670	12/2004	Bright	N/A	N/A
6971650	12/2004	Marelja	N/A	N/A
7287797	12/2006	Belloso	296/193.07	B62D 21/03
7651153	12/2009	Martin	180/65.6	B60G 17/017
7984941	12/2010	Lorenzo et al.	N/A	N/A
8127879	12/2011	Constans	180/89.12	B62D 33/06

8240748	12/2011	Chapman	N/A	N/A
8641133	12/2013	Scaringe et al.	N/A	N/A
8814209	12/2013	Way	N/A	N/A
8991900	12/2014	Yamaji et al.	N/A	N/A
9266569	12/2015	Tew	N/A	B62D 33/06
9272740	12/2015	Portenier	N/A	N/A
10207757	12/2018	Scaringe	N/A	N/A
10328863	12/2018	Spahn et al.	N/A	N/A
10479427	12/2018	Scaringe et al.	N/A	N/A
10800348	12/2019	Beet	N/A	N/A
10800463	12/2019	Beet	N/A	N/A
10829059	12/2019	Addison et al.	N/A	N/A
10967904	12/2020	Penz et al.	N/A	N/A
11161560	12/2020	Anderson et al.	N/A	N/A
11273770	12/2021	Cappuccio	N/A	N/A
11351922	12/2021	Estrada	N/A	N/A
11358655	12/2021	Urban	N/A	N/A
11485215	12/2021	Willison et al.	N/A	N/A
11505265	12/2021	Lee	N/A	B60L 50/66
2008/0003321	12/2007	Kerr et al.	N/A	N/A
2008/0191506	12/2007	Huotari et al.	N/A	N/A
2008/0231066	12/2007	Harrell	N/A	N/A
2010/0025132	12/2009	Hill et al.	N/A	N/A
2010/0101876	12/2009	Misencik	903/902	B62D 33/04
2010/0263954	12/2009	Constans	N/A	N/A
2011/0017527	12/2010	Oriet et al.	N/A	N/A
2012/0104793	12/2011	Danielson	180/65.21	B62D 29/046
2012/0104799	12/2011	Danielson et al.	N/A	N/A
2012/0104803	12/2011	Thomas	296/203.01	B62D 29/046
2013/0241237	12/2012	Dziuba et al.	N/A	N/A
2013/0341971	12/2012	Masini et al.	N/A	N/A
2015/0027795	12/2014	Hirai et al.	N/A	N/A
2018/0037151	12/2017	Bauer et al.	N/A	N/A
2018/0345777	12/2017	Birnschein et al.	N/A	N/A
2019/0283810	12/2018	Nothdurfter	N/A	B62D 23/00
2020/0369334	12/2019	Lee	N/A	N/A
2021/0253039	12/2020	Cappuccio	N/A	B60R 9/065
2022/0072940	12/2021	Hewitt et al.	N/A	N/A
2022/0080903	12/2021	Ano	N/A	N/A
2022/0097615	12/2021	Richter	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103359174	12/2012	CN	N/A
113212566	12/2020	CN	N/A
102013019677	12/2014	DE	N/A
1538072	12/2004	EP	N/A
2552763	12/2012	EP	N/A

3174680	12/2016	EP	N/A
3689717	12/2019	EP	N/A
2698601	12/1993	FR	N/A
H09150472	12/1996	JP	N/A
2007099186	12/2006	JP	N/A
2016005955	12/2015	JP	N/A
2019077960	12/2018	JP	N/A
2011145129	12/2010	WO	N/A
2016016662	12/2015	WO	N/A
2017135806	12/2016	WO	N/A
2018115827	12/2017	WO	N/A
2021142001	12/2020	WO	N/A
2023043650	12/2022	WO	N/A

OTHER PUBLICATIONS

International Search Report; International Searching Authority; International Patent Application No. PCT/US2021/012327; Mar. 9, 2021; 2 pages. cited by applicant

Written Opinion of the International Searching Authority; International Searching Authority; International Patent Application No. PCT/US2021/012327; Mar. 9, 2021; 8 pages. cited by applicant

International Search Report; International Searching Authority; International Patent Application No. PCT/US2021/012330; Mar. 9, 2021; 2 pages. cited by applicant

Written Opinion of the International Searching Authority; International Searching Authority; International Patent Application No. PCT/US2021/012330; Mar. 9, 2021; 9 pages. cited by applicant

International Search Report; International Searching Authority; International Patent Application No. PCT/US2021/012332; Mar. 24, 2021; 2 pages. cited by applicant

Written Opinion of the International Searching Authority; International Searching Authority; International Patent Application No. PCT/US2021/012332; Mar. 24, 2021; 7 pages. cited by applicant

Supplementary European Search Report; European Patent Office; European Application No. 21710384.5; Oct. 21, 2021; 12 pages. cited by applicant

Canadian Office Action; Canadian Intellectual Property Office; Canadian Patent Application No. 3,108,852; Apr. 13, 2022; 8 pages. cited by applicant

Communication pursuant to Article 94(3) EPC; European Patent Office; European Application No. 2170384.5; Apr. 3, 2023; 9 pages. cited by applicant

International Search Report; International Searching Authority; International Patent Application No. PCT/US2023/022426; Jun. 13, 2023; 2 pages. cited by applicant

Written Opinion of the International Searching Authority; International Searching Authority; International Patent Application No. PCT/US2023/022426; Jun. 13, 2023; 3 pages. cited by applicant

Canadian Office Action; Canadian Intellectual Property Office; Canadian Patent Application No. 3,108,852; Oct. 26, 2023; 5 pages. cited by applicant

Canadian Office Action; Canadian Intellectual Property Office; Canadian Patent Application No. 3,178,524; Mar. 20, 2024; 4 pages. cited by applicant

International Preliminary Report on Patentability; International Searching Authority; International Patent Application No. PCT/US2023/022426; Nov. 7, 2024; 4 pages. cited by applicant

Extended European Search Report, European Patent Office, European Patent Application No. 23808197.0, Feb. 4, 2025, 11 pages. cited by applicant

Canadian Office Action, Canadian Intellectual Property Office, Canadian Patent Application No.

3,178,524, Feb. 3, 2025, 4 pages. cited by applicant
Communication pursuant to Article 94(3) EPC, European Patent Office, European Patent Application No. 21710384.5, Mar. 26, 2025, 10 pages. cited by applicant
Japanese Office Action, Japan Patent Office, Japanese Patent Application No. 2022-541654, May 14, 2025, 4 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) The present application is a continuation of, and claims priority to, U.S. application Ser. No. 18/429,814, which was filed on Feb. 1, 2024, which is a continuation of U.S. application Ser. No. 18/125,805, which was filed on Mar. 24, 2023 and which issued as U.S. Pat. No. 11,891,120, and which is a continuation of U.S. application Ser. No. 17/962,143, which was filed on Oct. 7, 2022 and which issued as U.S. Pat. No. 11,613,310, and which is a continuation of U.S. application Ser. No. 17/745,434, which was filed on May 16, 2022 and which issued as U.S. Pat. No. 11,472,485, and which is a continuation-in-part application of, and claims the priority benefit of, U.S. application Ser. No. 17/716,687, which was filed on Apr. 8, 2022 and which issued as U.S. Pat. No. 11,472,484, and which is a continuation of U.S. application Ser. No. 17/552,718, which issued as U.S. Pat. No. 11,299,208 on Apr. 12, 2022, and which is a continuation application of U.S. application Ser. No. 17/142,766, which issued as U.S. Pat. No. 11,220,297 on Jan. 11, 2022, and which claims priority to, and the benefit of, U.S. Provisional App. Ser. No. 62/957,577, which was filed on Jan. 6, 2020. The contents of those applications are incorporated by reference herein in their entireties.

TECHNICAL FIELD

(1) The present disclosure generally relates to land vehicles and methods of making land vehicles, and, more particularly, to utility and delivery vehicles and methods of making utility and delivery vehicles.

BACKGROUND

(2) Current systems and methods for manufacturing utility and delivery vehicles suffer from a variety of drawbacks and limitations. For those reasons, among others, there remains a need for further improvements in this technological field.

SUMMARY

(3) The present disclosure may comprise one or more of the following features and combinations thereof.

(4) According to one aspect of the present disclosure, an electric vehicle may include a front cage, a rear floor, an intermediate section, a utility cabinet, and a flatbed. The front cage may at least partially define an operator cabin. The rear floor may be positioned rearward of the front cage in a longitudinal direction. The intermediate section may be disposed at least partially between the front cage and the rear floor in the longitudinal direction. The utility cabinet may include a plurality of drawers accessible from a utility space located outside the operator cabin. The flatbed may be at least partially defined by the rear floor. The flatbed may be open to the ambient environment and define the utility space.

(5) In some embodiments, the utility cabinet may be arranged at least partially between the front cage and the rear floor in the longitudinal direction, and when one or more of the plurality of drawers is in a closed state, the one or more of the plurality of drawers in the closed state may be

aligned with the intermediate section in the longitudinal direction. When one or more of the plurality of drawers is in an opened state, the one or more of the plurality of drawers in the opened state may extend rearward in the longitudinal direction into the utility space.

(6) In some embodiments, the utility cabinet may extend forward in the longitudinal direction to the front cage. The utility cabinet may extend at least partially into the operator cabin. Additionally, in some embodiments, the utility cabinet may directly contact a rearwardly-facing exterior wall of the front cage that is interconnected with a roof of the front cage, and the exterior wall, the roof, and an uppermost surface of the utility cabinet in a vertical direction may cooperate to define a storage space accessible from the utility space.

(7) In some embodiments, the utility cabinet may be arranged at least partially between the front cage and the rear floor in the longitudinal direction, the intermediate section may include an alcove aligned with the utility cabinet in the longitudinal direction, the alcove may define an opening into the intermediate section in a lateral direction perpendicular to the longitudinal direction, and the opening may be closed off from the operator cabin. Additionally, in some embodiments, the flatbed may include a bench arranged in the utility space that is supported by the rear floor, and the plurality of drawers of the utility cabinet may be disposed above the bench in a vertical direction to permit opening and closing of the plurality of drawers without interference with the bench. In some embodiments still, movement of the vehicle may be driven by one or more electric motors, the vehicle may include a monocoque that at least partially defines the front cage, the rear floor, the intermediate section, and the flatbed, and the monocoque may be formed from composite materials.

(8) In some embodiments, the flatbed may include (i) first sidewall that extends upwardly away from the rear floor in a vertical direction, (ii) a second sidewall arranged opposite the first sidewall that extends upwardly away from the rear floor in the vertical direction, (iii) a rear gate that extends between the first sidewall and the second sidewall in a lateral direction perpendicular to the longitudinal direction to at least partially close off the utility space, (iv) a first guide rail interconnected with the first sidewall that extends upwardly away from the first sidewall in the vertical direction to define a first slot between the first sidewall and the first guide rail, and (v) a second guide rail interconnected with the second sidewall that extends upwardly away from the second sidewall in the vertical direction to define a second slot between the second sidewall and the second guide rail. The flatbed may include (vi) a rear bumper extending rearwardly in the longitudinal direction away from the rear gate and (vii) a vice mounted to the rear bumper.

(9) According to another aspect of the present disclosure, an electric vehicle may include a front cage, a rear floor, an intermediate section, and a flatbed. The front cage may at least partially define an operator cabin having a roof. The rear floor may be positioned rearward of the front cage in a longitudinal direction. The intermediate section may be disposed at least partially between the front cage and the rear floor in the longitudinal direction. The flatbed may be at least partially defined by the rear floor. The flatbed may be open to the ambient environment and define a utility space accessible from a rear end of the vehicle. The flatbed may include a first sidewall that extends upwardly away from the rear floor in a vertical direction to at least partially define the utility space and a second sidewall arranged opposite the first sidewall that extends upwardly away from the rear floor in the vertical direction to at least partially define the utility space. The intermediate section may include a reinforcement beam that is interconnected with the roof at one end thereof and interconnected with one of the first sidewall and the second sidewall at another end thereof opposite the one end.

(10) In some embodiments, the reinforcement beam may extend oblique to the one of the first sidewall and the second sidewall to define an obtuse angle between the reinforcement beam and the one of the first sidewall and the second sidewall. The reinforcement beam may at least partially define an alcove of the intermediate section that is disposed at least partially between the front cage and the rear floor in the longitudinal direction, and the alcove may define an opening into the intermediate section in a lateral direction perpendicular to the longitudinal direction. The flatbed

may include a first guide rail interconnected with the first sidewall that extends upwardly away from the first sidewall in the vertical direction to define a first slot between the first sidewall and the first guide rail and a second guide rail interconnected with the second sidewall that extends upwardly away from the second sidewall in the vertical direction to define a second slot between the second sidewall and the second guide rail, and at least one of the first guide rail and the second guide rail may be directly interconnected with the reinforcement beam. The flatbed may include a rear gate that extends between the first sidewall and the second sidewall in a lateral direction perpendicular to the longitudinal direction to at least partially close off the utility space, a rear bumper extending rearwardly in the longitudinal direction away from the rear gate, and a vice mounted to the rear bumper.

(11) In some embodiments, the electric vehicle may include a utility cabinet including a plurality of drawers accessible from the utility space, the utility cabinet may be arranged at least partially between the front cage and the rear floor in the longitudinal direction, and the utility cabinet may extend forward in the longitudinal direction to the front cage. The utility cabinet may extend at least partially into the operator cabin. The utility cabinet may directly contact a rearwardly-facing exterior wall of the front cage that is interconnected with the roof of the front cage, and the exterior wall, the roof, and an uppermost surface of the utility cabinet in the vertical direction may cooperate to define a storage space accessible from the utility space.

(12) According to yet another aspect of the present disclosure, an electric vehicle may include a front cage, a rear floor, an intermediate section, a utility cabinet, and a flatbed. The front cage may at least partially define an operator cabin having a roof. The rear floor may be positioned rearward of the front cage in a longitudinal direction. The intermediate section may be disposed at least partially between the front cage and the rear floor in the longitudinal direction. The utility cabinet may include a plurality of drawers accessible from a utility space located outside the operator cabin. The flatbed may be at least partially defined by the rear floor. The flatbed may be open to the ambient environment and define the utility space. The flatbed may include a first sidewall that extends upwardly away from the rear floor in a vertical direction to at least partially define the utility space and a second sidewall arranged opposite the first sidewall that extends upwardly away from the rear floor in the vertical direction to at least partially define the utility space. The intermediate section may include a reinforcement beam that is interconnected with the roof at one end thereof and interconnected with one of the first sidewall and the second sidewall at another end thereof opposite the one end. The reinforcement beam may extend oblique to the one of the first sidewall and the second sidewall to define an obtuse angle between the reinforcement beam and the one of the first sidewall and the second sidewall.

(13) These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The invention described herein is illustrated by way of example and not by way of limitation in the accompanying figures. For simplicity and clarity of illustration, elements illustrated in the figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements for clarity. Further, where considered appropriate, reference labels have been repeated among the figures to indicate corresponding or analogous elements.

(2) FIG. 1 depicts side elevation views of a number of electric vehicles that may be included in an electric vehicle line according to certain embodiments of the disclosure;

(3) FIG. 2 is a perspective view of a monocoque or unibody that may be incorporated into any electric vehicle of the disclosure;

- (4) FIG. 3 is a partially exploded assembly view of an electric vehicle according to at least one embodiment of the disclosure;
- (5) FIG. 4 is a partial schematic rear end view of a conventional delivery vehicle;
- (6) FIG. 5 is a partial schematic rear end view of a delivery vehicle according to at least one embodiment of the disclosure;
- (7) FIG. 6 is a table illustrating United States standard vehicle classes by gross vehicular weight rating (GVWR);
- (8) FIG. 7 is a partial schematic depiction of a composite structure that may be used to form a monocoque or unibody of any electric vehicle of the disclosure;
- (9) FIG. 8 is a diagrammatic depiction of at least one modular mold system according to certain embodiments of the disclosure;
- (10) FIG. 9 is a perspective view of a monocoque system formed from a number of mold units included in the at least one modular mold system of FIG. 8;
- (11) FIG. 10 is a simplified flowchart of a method of forming a monocoque of an electric vehicle using one modular mold system according to one embodiment of the disclosure;
- (12) FIG. 11 is a simplified flowchart of one portion of another method of forming a monocoque of an electric vehicle using one modular mold system according to another embodiment of the disclosure;
- (13) FIG. 12 is a diagrammatic view of another portion of the method of FIG. 11;
- (14) FIG. 13 is a simplified flowchart of a method of forming multiple monocoques of electric vehicles using at least one modular mold system according to yet another embodiment of the disclosure;
- (15) FIG. 14 is a rear perspective view of one of the electric vehicles depicted in FIG. 1; and
- (16) FIG. 15 is a side elevation view of another one of the electric vehicles depicted in FIG. 1.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(17) While the concepts of the present disclosure are susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and will be described herein in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives consistent with the present disclosure and the appended claims.

(18) References in the specification to “one embodiment,” “an embodiment,” “an illustrative embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may or may not necessarily include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to effect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. Additionally, it should be appreciated that items included in a list in the form of “at least one A, B, and C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C). Similarly, items listed in the form of “at least one of A, B, or C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C).

(19) In the drawings, some structural or method features, such as those representing devices, modules, instructions blocks and data elements, may be shown in specific arrangements and/or orderings for ease of description. However, it should be appreciated that such specific arrangements and/or orderings may not be required. Rather, in some embodiments, such features may be arranged in a different manner and/or order than shown in the illustrative figures. Additionally, the inclusion of a structural or method feature in a particular figure is not meant to imply that such feature is required in all embodiments and, in some embodiments, may not be included or may be combined with other features.

(20) In some embodiments, schematic elements used to represent blocks of a method may be manually performed by a user. In other embodiments, implementation of those schematic elements may be automated using any suitable form of machine-readable instruction, such as software or firmware applications, programs, functions, modules, routines, processes, procedures, plug-ins, applets, widgets, code fragments and/or others, for example, and each such instruction may be implemented using any suitable programming language, library, application programming interface (API), and/or other software development tools. For instance, in some embodiments, the schematic elements may be implemented using Java, C++, and/or other programming languages. Similarly, schematic elements used to represent data or information may be implemented using any suitable electronic arrangement or structure, such as a register, data store, table, record, array, index, hash, map, tree, list, graph, file (of any file type), folder, directory, database, and/or others, for example.

(21) Further, in the drawings, where connecting elements, such as solid or dashed lines or arrows, are used to illustrate a connection, relationship, or association between or among two or more other schematic elements, the absence of any such connection elements is not meant to imply that no connection, relationship, or association can exist. In other words, some connections, relationships, or associations between elements may not be shown in the drawings so as not to obscure the disclosure. In addition, for ease of illustration, a single connecting element may be used to represent multiple connections, relationships, or associations between elements. For example, where a connecting element represents a communication of signals, data or instructions, it should be understood by those skilled in the art that such element may represent one or multiple signal paths (e.g., a bus), as may be needed, to effect the communication.

(22) Referring now to FIG. 1, an illustrative line **100** of land vehicles includes a plurality of land vehicles. In the illustrative embodiment, the land vehicle line **100** includes, but is not limited to, a two-passenger flatbed utility vehicle **110**, a 650 cubic foot capacity delivery vehicle **120**, a 1000 cubic foot capacity delivery vehicle **130**, a six-passenger flatbed utility vehicle **140**, and a 1200 cubic foot capacity delivery vehicle **150**. However, in some embodiments, the land vehicle line **100** may include any vehicle having a capacity within a particular range, such as a range of from 400 cubic feet to 1400 cubic feet, for example. In keeping with industry terminology, the phrase “cubic foot capacity” may be shortened or abbreviated herein to simply “cube.” It should be appreciated that the phrase “cubic foot capacity” as contemplated herein may refer to a storage volume or storage capacity of a particular land vehicle. In any case, as will be apparent from the discussion that follows, one or more vehicles of the vehicle line **100** may be manufactured using the systems and methods described herein.

(23) In the illustrative embodiment, each of the vehicles included in the vehicle line **100** (i.e., each of the vehicles **110**, **120**, **130**, **140**, **150**) includes a monocoque or unibody **200** (see FIG. 2) supporting wheels (e.g., wheels **112**, **122**, **132**, **142**, **152**) to permit movement of the particular vehicle relative to an underlying surface in use thereof. As described herein, the monocoque **200** is a single-piece, monolithic structure unsupported by an internal chassis. The monocoque **200** includes a front cage **210** defining an operator cabin **212** and a rear floor **220** positioned rearward of the front cage **210**. The monocoque **200** illustratively has a composite construction (e.g., the composite structure **700** shown in FIG. 7) such that each of the front cage **210** and the rear floor **220** are formed from one or more composite materials, as described in greater detail below.

(24) At least some of the vehicles (e.g., the vehicles **110**, **140**) of the illustrative line **100** may be embodied as, included in, or otherwise adapted for use with, electric utility vehicles. Furthermore, at least some of the vehicles (e.g., the vehicles **120**, **130**, **150**) of the illustrative line **100** may be embodied as, included in, or otherwise adapted for use with, electric vehicles having enclosed stowage compartments. Of course, in other embodiments, it should be appreciated that the vehicles of the line **100** may be embodied as, included in, or otherwise adapted for use with, other suitable vehicles.

(25) It should be appreciated each of the vehicles of the illustrative line **100** may be employed in a

variety of applications. In some embodiments, one or more vehicles of the line **100** may be embodied as, or otherwise included in, a fire and emergency vehicle, a refuse vehicle, a coach vehicle, a recreational vehicle or motorhome, a municipal and/or service vehicle, an agricultural vehicle, a mining vehicle, a specialty vehicle, an energy vehicle, a defense vehicle, a port service vehicle, a construction vehicle, and a transit and/or bus vehicle, just to name a few. Additionally, in some embodiments, one or more vehicles of the line **100** may be adapted for use with, or otherwise incorporated into, tractors, front end loaders, scraper systems, cutters and shredders, hay and forage equipment, planting equipment, seeding equipment, sprayers and applicators, tillage equipment, utility vehicles, mowers, dump trucks, backhoes, track loaders, crawler loaders, dozers, excavators, motor graders, skid steers, tractor loaders, wheel loaders, rakes, aerators, skidders, bunchers, forwarders, harvesters, swing machines, knuckleboom loaders, diesel engines, axles, planetary gear drives, pump drives, transmissions, generators, and marine engines, among other suitable equipment.

(26) In the illustrative embodiment, each of the vehicles of the line **100** includes one or more electric motors (not shown) capable of generating rotational power that may be transmitted to the wheels to drive movement of the vehicle. As such, each of the illustrative vehicles is embodied as, or otherwise includes, an electric vehicle. Details regarding the electric motor(s) included in each vehicle and associated powertrain and/or suspension components are described in U.S. patent application Ser. No. 17/142,814, the contents of which are incorporated herein by reference in their entirety.

(27) Each of the vehicles of the illustrative line **100** does not include an internal combustion engine or powerplant, at least in some embodiments. Furthermore, each of the vehicles of the illustrative line **100** does not include an engine or powerplant housed by the front cage **210** and positioned above an underside **214** of the monocoque **200**. Instead, as described in U.S. patent application Ser. No. 17/142,814, multiple electric motors or powerplants are removably coupled to the underside **214** of the monocoque **200** of each vehicle of the illustrative line **100**.

(28) It should be appreciated that the vehicles of the illustrative vehicle line **100** may each include one or more features that improve the experience of the driver, the owner, and/or maintenance personnel. Such features may include, but are not limited to, a low floor, a modular battery system, air springs and/or air ride features, an independent rear suspension, an independent front suspension, thermal battery management capability, flexible shelving options, desirable driver sightlines, LED lighting, telematics/driver feedback, features to facilitate maintenance, an aerodynamic body, and advanced safety systems. Further details regarding at least some of these features are provided herein.

(29) Referring now to FIG. 2, in addition to the front cage **210** and the rear floor **220**, at least in some embodiments, the monocoque **200** includes an intermediate section **230** arranged between the front cage **210** and the rear floor **220**. The intermediate section **230** may form a portion of a floor section arranged forward of the rear floor **220**. As described in greater detail below with reference to FIG. 8, each of the front cage **210**, the rear floor **220**, and the intermediate section **230** may be associated with, and formed with, a corresponding mold unit of a modular mold system (e.g., the system **800**). Furthermore, as described in greater detail below with reference to FIG. 9, the mold units of the modular mold system may be joined together to form a monocoque mold (e.g., the monocoque mold **900**) into which composite materials may be introduced to form the monocoque **200**.

(30) In the illustrative embodiment, the monocoque **200** combines what would traditionally be formed from one or more separate structures (e.g., one or more body components and one or more frame components) into a single-piece, monolithic structure. As such, any vehicle of the present disclosure incorporating the monocoque **200** does not include an internal chassis or frame structure that supports separate body components (e.g., panels, doors, etc.). Due at least in part to consolidation of body and frame components into an integrally-formed structure, the illustrative

monocoque **200** may be associated with, or otherwise facilitate, improved manufacturability and/or simplified maintenance compared to other configurations.

(31) Depending on the particular vehicle type and monocoque configuration, one or more dimensions of the intermediate section **230** of the monocoque **200** may vary. In one example, the intermediate section **230** may have a first length associated with, and defined by, a small intermediate section mold unit (e.g., the mold unit **832** shown in FIG. **8**). In that example, the first length of the intermediate section **230** may at least partially define a stowage compartment of a 650 cubic feet delivery vehicle (e.g., the vehicle **120**). In another example, the intermediate section **230** may have a second length associated with, and defined by, a medium intermediate section mold unit (e.g., the mold unit **834** shown in FIG. **8**). In that example, the second length of the intermediate section **230** may at least partially define a stowage compartment of a 1000 cubic feet delivery vehicle (e.g., the vehicle **130**). In yet another example, the intermediate section **230** may have a third length associated with, and defined by, a large intermediate section mold unit (e.g., the mold unit **836** shown in FIG. **8**). In that example, the third length of the intermediate section **230** may at least partially define a stowage compartment of a 1200 cubic feet delivery vehicle (e.g., the vehicle **150**).

(32) Furthermore, depending on the particular vehicle type and monocoque configuration, the intermediate section **230** of the monocoque **200** may be omitted entirely. In such embodiments, the front cage **210** and the rear floor **220** may be integrally-formed as a single-piece, monolithic structure without the intermediate section **230** interposed therebetween. It should be appreciated that the utility vehicles **110** and **140** may each include a monocoque formed without the intermediate section **230**, at least in some embodiments.

(33) Referring now to FIG. **3**, a vehicle **300** incorporates the monocoque **200** with the intermediate section **230** arranged between the front cage **210** and the rear floor **220**. Additionally, the vehicle **300** includes a cab hood **302** arranged above the front cage **210** to enclose the operator cabin **212** and a stowage compartment **310** arranged rearward of the front cage **210** and the cab hood **302**. In the illustrative embodiment, the stowage compartment **310** is at least partially defined by the intermediate section **230** and the rear floor **220** and has a roof **312** and sidewalls **314**. The illustrative vehicle **300** may be similar to any one of the vehicles **120**, **130**, **150** discussed above, at least in some embodiments.

(34) Because the monocoque **200** has a composite construction as indicated above, it should be appreciated that any vehicle described herein that incorporates the monocoque **200** (e.g., any of the vehicles **110**, **120**, **130**, **140**, **150**, **300**, **500**) incorporates a composite structure (e.g., the structure **700** shown in FIG. **7**). In the case of the vehicle **300**, each of the intermediate section **230**, the roof **312**, and the sidewalls **314** is formed from composite materials and has a composite structure, at least in some embodiments. In those embodiments, each of the intermediate section **230**, the roof **312**, and the sidewalls **314** does not include metallic material.

(35) Referring now to FIG. **4**, a prior art delivery vehicle **400** includes a stowage compartment **410**. The stowage compartment **410** includes a floor **412**, a pair of sidewalls **414**, a ceiling **416**, and a refrigeration unit **418** at least partially housed by the stowage compartment **410** and configured to cool the stowage compartment **410**. The rear end of the vehicle **400** includes a landing **404** and a step **406** that leads to the floor **412** of the stowage compartment **410**.

(36) As depicted in FIG. **4**, the landing **404** has a landing height **424** above ground level **402** and the step **406** has a step height **426** above the landing **404**. The floor **412** has a floor height **422** above the ground level **402** that includes both the landing height **424** and the step height **426**. Typically, the landing height **424** is about 25 inches, the step height **426** is about ten inches, and the floor height **422** is about 35 inches.

(37) Referring now to FIG. **5**, a delivery vehicle **500** may include a monocoque (e.g., the monocoque **200**) described above with reference to FIG. **2**. Furthermore, in some embodiments, the vehicle **500** may be similar to one or more of the vehicles **120**, **130**, **150** described above. In any

case, the illustrative delivery vehicle **500** includes a stowage compartment **510** having a floor **512**, a pair of sidewalls **514**, and a ceiling **516**, as well as a refrigeration unit **518** housed by the stowage compartment **510**. Unlike the prior art delivery vehicle **400**, however, the vehicle **500** lacks a step corresponding to the step **406**. As such, the floor **512** has a floor height **522** that substantially corresponds to, and may be equal to, the landing height **424**. The floor height **522** may be less than thirty inches, such as in the range of 22 to 28 inches, for example. A pair of wheel wells **530** formed within the stowage compartment **510** are offset from one another by a separation distance **532**. In certain embodiments, the separation distance **532** may be about 50 inches.

(38) In some cases, the prior art delivery vehicle **400** suffers from one or more disadvantages not associated with the illustrative vehicle **500**. In one respect, the sidewalls **414** and the ceiling **416** of the prior art vehicle **400** are typically formed of metallic material such as aluminum, for example, which is a poor thermal insulator. As such, the compartment **410** may be poorly insulated and have a tendency to adopt the temperature of the ambient environment relatively quickly. That may be especially the case in the summer when radiant heat from the sun supplements the ambient hot air to exacerbate the warming of the compartment **410**. In contrast, the sidewalls **514** and the ceiling **516** of the illustrative vehicle **500** are formed of composite materials, which exhibit superior insulating characteristics compared to metallic material such as aluminum. Accordingly, the compartment **510** is insulated from the ambient environment to a greater degree than the compartment **410**. That insulation may be particularly advantageous in cases in which the vehicle **500** is a refrigerated vehicle such as a food delivery vehicle, for instance. It should be appreciated that the insulating properties of the compartment **510** may ease the cooling burden on the refrigeration unit **518** and thereby increase performance of the refrigeration unit **518**. Additionally, in certain circumstances, increased performance of the refrigeration unit **518** may enable the vehicle **500** to be provided with a smaller refrigeration unit **518** than would typically be required by the prior art vehicle **400**.

(39) Another drawback associated with the prior art vehicle **400** is the elevated nature of the floor **412** relative to the ground level **402**. It should be appreciated that the elevated floor **412** is not merely a design choice but rather a feature often necessitated to accommodate inclusion of the internal chassis or frame, the powertrain, and associated components. Put another way, to accommodate the mounting of a conventional internal combustion engine and other powertrain components (e.g., a transmission, transaxle, and/or a differential) to an internal chassis, the floor **412** is elevated above the ground level **402** by the floor height **422**. Consequently, the elevated floor **412** reduces the storage capacity and/or volume of the stowage compartment **410** and requires the provision of the step **406**. Delivery personnel using the vehicle **400** must therefore step up onto the landing **404** and ascend the step **406** in order to access the compartment **410**.

(40) The illustrative vehicle **500** obviates a number of the aforementioned disadvantages by eliminating the necessity of the elevated floor **412**. Due in part to the provision of the monocoque **200** as a single-piece, monolithically formed structure having a relatively lightweight composite construction, and due in part to the absence of powertrain components typically provided in other configurations (e.g., a central drive shaft beneath the underside **214** of the monocoque **200** that provides a rotational input to a differential), the floor **512** need not be elevated above the ground level like the floor **412**. As a result, the vehicle **500** allows increased stowage capacity of the stowage compartment **510** to be achieved without raising the ceiling **516**. Moreover, because a step similar to the step **406** may be omitted from the vehicle **500**, the floor height **522** corresponds to the landing height **424** of the conventional vehicle **400**, and delivery personnel may avoid the effort of ascending both the landing **404** and the step **406** to access the stowage compartment **510** of the vehicle **500**. Notably, it should be appreciated that a rear bumper of the vehicle **500** may be slightly lower than the floor **512** and that delivery personnel may access the compartment **510** by first stepping on the rear bumper. In some embodiments, the rear bumper may have a height of about 20 inches above the ground level, whereas the floor **512** may have a height of about 25 inches above

the ground level.

(41) Referring now to FIG. 6, in the United States, trucks are often classified according to their gross vehicular weight rating (GVWR). Those truck classifications, the associated duty classifications, and the corresponding GVWRs are illustrated in the table 600. In the illustrative embodiment, one or more of the vehicles 110, 120, 130, 140, 150 has a GVWR (i.e., accounting for the weight of the truck when empty and the payload carrying capacity of the truck when full) of between 6,000 pounds and 19,800 pounds. In some embodiments, one or more of the vehicles 110, 120, 130, 140, 150 has a GVWR of between 10,001 pounds and 14,000 pounds such that one or more of the vehicles 110, 120, 130, 140, 150 is embodied as, or otherwise includes, a Class 3 truck. In one particular example, in some embodiments, the 1000 cubic foot capacity vehicle 130 weighs roughly 6,500 pounds when empty and has a 6,000 pound payload capacity such that the vehicle 130 has a GVWR of about 12,500 pounds. Of course, it should be appreciated that in other embodiments, the vehicle line 100 may include one or more vehicles in Class 3, one or more vehicles in Class 4, and/or one or more vehicles in Class 5.

(42) In some embodiments, the systems and methods described herein may find particular utility in connection with delivery vehicles in Classes 3 through 5. For example, the methods 1000, 1100, 1300 described below may be utilized to form a monocoque for a delivery vehicle having a GVWR between 10,001 pounds and 19,500 pounds. The stowage capacity of such a vehicle may be between 450 cubic feet and 1200 cubic feet. In certain embodiments, the stowage compartment (e.g., the compartment 510) of the vehicle may be isolated from the operator cabin (e.g., the operator cabin 212) of the vehicle.

(43) Referring now to FIG. 7, any vehicle of the present disclosure includes a monocoque having the composite structure 700. In the illustrative embodiment, the composite structure 700 incorporates one or more relatively lightweight, low-density materials to impart a relatively lightweight construction to the vehicle. As discussed below, the illustrative composite structure 700 includes one or more of the following: balsa wood, plastic, fiberglass, resin, Kevlar, honeycomb, and carbon fiber. The composite structure 700 does not include, and is not formed from, metallic material, at least in some embodiments. In those embodiments, the monocoque (e.g., the monocoque 200) incorporating the composite structure 700 does not include metallic material.

(44) The illustrative composite structure 700 includes a core 702 and a shell 704 that at least partially surrounds the core 702. In the illustrative embodiment, the core 702 is formed from balsa wood and/or one or more of the following composite, non-metallic materials: unidirectional fiberglass, multi-directional fiberglass, Kevlar, carbon fiber, plastic, honeycomb, or other suitable composite, non-metallic materials. Of course, in other embodiments, the core 702 may be formed from other suitable materials to provide a relatively lightweight construction to the composite structure 700. The illustrative shell 704 is formed from fiberglass and resin. However, in other embodiments, the shell 704 may be formed from other suitable materials. Additionally, in the illustrative embodiment, the composite structure 700 includes a laminate layer 706 that at least partially covers the shell 704.

(45) It should be appreciated that the composite structure 700 used to form the monocoque of any vehicle of the present disclosure offers a number of advantages over multi-piece metallic constructions of conventional vehicles. In one respect, the single-piece monolithic structure formed with the composite structure 700 has fewer parts and offers greater structural simplicity than vehicle constructions requiring multiple parts. In another respect, the structural simplicity afforded by the composite structure 700 may facilitate maintenance and improve structural efficiency. In yet another respect, due to a lack of metallic material, the composite structure 700 may minimize or eliminate rust and/or corrosion and thereby have a service life that exceeds the service life of vehicles having conventional constructions. In some instances, monocoques incorporating composite structures 700 consistent with the teachings of the present disclosure may have service lives of 20 years or more.

(46) Referring now to FIGS. 8 and 9, a modular mold system **800** (see FIG. 8) includes a number of illustrative mold units that may be selected and arranged to form a monocoque system **900** (see FIG. 9). It should be appreciated that when arranged to form the monocoque system **900**, the selected mold units of the modular system **800** are utilized to form a monocoque such as the above-described monocoque **200**, for example. Furthermore, it should be appreciated that similar reference numerals in the 800 and 900 series are used to designate corresponding features of the modular mold system **800** and the monocoque system **900**.

(47) The illustrative mold system **800** includes a front cage mold unit **810**, a rear floor mold unit **820**, and a plurality of intermediate mold units **830** having a small intermediate section mold unit **832**, a medium intermediate section mold unit **834**, and a large intermediate section mold unit **836**. As discussed below, each of the mold units **810**, **820**, **832**, **834**, **836** has a mold cavity having a size and a shape corresponding to a corresponding feature of the monocoque system **900** such that subsequent to introduction of the composite materials (e.g., the materials of the composite structure **700**) into the mold cavity, the corresponding feature of the monocoque system **900** will be formed. Accordingly, the front cage mold unit **810** includes a front cage mold cavity **912** that has a size and a shape corresponding to the front cage **910** of the monocoque system **900** (and also the front cage **210**). The rear floor mold unit **820** includes a rear floor mold cavity **922** that has a size and a shape corresponding to the rear floor **920** of the monocoque system **900** (and also the rear floor **220**). The intermediate mold units **832**, **834**, **836** include respective intermediate mold cavities **933**, **935**, **937** each having a size and a shape corresponding to the respective intermediate section **932**, **934**, **936** of the monocoque system **900** (and also the intermediate section **230**).

(48) As evident from FIGS. 8 and 9, each of the intermediate mold units **832**, **834**, **836** is sized for positioning between the front cage mold unit **810** and the rear floor mold unit **820** to form the monocoque system **900**. It should be appreciated that any one of the intermediate mold units **832**, **834**, **836** may be selected and arranged between the front cage mold unit **810** and the rear floor mold unit **820** to form the monocoque system **900**. Selection of the particular mold unit **832**, **834**, **836** is based on the configuration of the vehicle and the monocoque included therein, as further discussed below.

(49) In the illustrative embodiment, the front cage mold cavity **912** of the front cage mold unit **810** has an opening **914** at a rear end thereof (i.e., the end closest to one of the intermediate sections **932**, **934**, **936** as shown in FIG. 9) to establish a fluidic coupling between the cavity **912** and another component of the mold system **800**. In some embodiments, a fluidic coupling may be established between the front cage mold cavity **912** and one of the intermediate mold cavities **933**, **935**, **937** when the front cage mold unit **810** is arranged contiguously with one of the corresponding intermediate mold units **832**, **834**, **836**. Additionally, in some embodiments, a fluidic coupling may be established between the front cage mold cavity **912** and the rear floor mold cavity **922** when the front cage mold unit **810** is arranged contiguously with the rear floor mold unit **820**.

(50) In the illustrative embodiment, the rear floor mold cavity **922** of the rear floor mold unit **820** has an opening **924** at a front end thereof (i.e., the end closest to one of the intermediate sections **932**, **934**, **936** as shown in FIG. 9) to establish a fluidic coupling between the cavity **922** and another component of the mold system **800**. Each of the intermediate mold cavities **933**, **935**, **937** of the intermediate mold units **832**, **834**, **836** has an opening **938** at a front end thereof (i.e., the end closest to the front cage **910** as shown in FIG. 9) and an opening **940** at a rear end thereof (i.e., the end closest to the rear floor **920** as shown in FIG. 9). When one of the intermediate mold units **832**, **834**, **836** is arranged contiguously with the front cage mold unit **810**, a fluidic coupling is established between the corresponding intermediate mold cavity **933**, **935**, **937** and the front cage mold cavity **912** via the openings **914**, **938**. Additionally, when one of the intermediate mold units **832**, **834**, **836** is arranged contiguously with the rear floor mold unit **820**, a fluidic coupling is established between the corresponding intermediate mold cavity **933**, **935**, **937** and the rear floor mold cavity **922** via the openings **924**, **940**.

(51) It should be appreciated that the front end of each of the illustrative intermediate mold units **832**, **834**, **836** is configured for direct connection and attachment to the rear end of the front cage mold unit **810**. Furthermore, it should be appreciated that the rear end of each of the illustrative intermediate mold units **832**, **834**, **836** is configured for direct connection and attachment to the front end of the rear floor mold unit **820**. Consequently, when any one of the intermediate mold units **832**, **834**, **836** is directly connected with the front cage mold unit **810** and the rear floor mold unit **820**, the front cage mold cavity **912**, the corresponding intermediate mold cavity **933**, **935**, **937**, and the rear floor mold cavity **922** are fluidically coupled to one another in a contiguous arrangement to establish a continuous monocoque mold cavity into which composite materials may be introduced to form the monocoque as a single-piece, monolithic structure.

(52) It should also be apparent that the rear end of the illustrative front cage mold unit **810** is configured for direct connection and attachment to the front end of the rear floor mold unit **820**. As a result, when the front cage mold unit **810** is directly connected to the rear floor mold unit **820**, the front cage mold unit **810** and the rear floor mold unit **820** are fluidically coupled to one another in a contiguous arrangement to establish a continuous monocoque mold cavity into which composite materials may be introduced to form the monocoque as a single-piece, monolithic structure.

(53) In the illustrative embodiment, the small intermediate section mold unit **832** has a length **L1** as suggested by FIG. **9**. The medium intermediate section mold unit **834** has a length **L2** that is greater than the length **L1**, at least in some embodiments. The large intermediate section mold unit **836** has a length **L3** that is greater than the length **L2** and the length **L1**, at least in some embodiments.

(54) In some embodiments, the small intermediate section mold unit **832** may be used to form the intermediate section **932** of the monocoque system **900** such that the monocoque at least partially produced using the mold unit **832** is included in a vehicle having a storage volume of 650 cubic feet (e.g., the vehicle **120**). Additionally, in some embodiments, the medium intermediate section mold unit **834** may be used to form the intermediate section **934** of the monocoque system **900** such that the monocoque at least partially produced using the mold unit **834** is included in a vehicle having a storage volume of 1000 cubic feet (e.g., the vehicle **130**). In some embodiments still, the large intermediate section mold unit **836** may be used to form the intermediate section **936** of the monocoque system **900** such that the monocoque at least partially produced using the mold unit **836** is included in a vehicle having a storage volume of 1200 cubic feet (e.g., the vehicle **150**).

(55) Referring now to FIG. **10**, an illustrative method **1000** of forming a monocoque (e.g., the monocoque **200**) using a modular mold system (e.g., the system **800**) is depicted. The method **1000** corresponds to, or is otherwise associated with, performance of the blocks described below in the illustrative sequence of FIG. **10**. It should be appreciated, however, that the method **1000** may be performed in one or more sequences different from the illustrative sequence. Furthermore, it should be appreciated that one or more of the blocks described below may be executed contemporaneously and/or in parallel with one another. In some embodiments, the method **1000** may be performed manually by one or more operators. In other embodiments, the method **1000** may be embodied as, or otherwise include, a set of instructions that are performed by an automated control system.

(56) The illustrative method **1000** begins with block **1002**. In block **1002**, the operator(s) or the control system selects a land vehicle type or a monocoque configuration for a particular land vehicle. It should be appreciated that to perform block **1002**, the operator(s) or the control system may select any vehicle envisioned by the present disclosure or any monocoque configuration associated with a particular vehicle contemplated by the present disclosure. From block **1002**, the method **1000** subsequently proceeds to block **1004**.

(57) In block **1004** of the illustrative method **1000**, the operator(s) or the control system selects a first mold unit of the modular mold system based upon the selected vehicle type or monocoque configuration. In the illustrative embodiment, to perform block **1004**, the operator(s) or the control system selects the rear floor mold unit **820** of the modular system **800** in block **1006**. However, in other embodiments, it should be appreciated that block **1004** may be performed by selecting (i) the

small intermediate section mold unit **832** (i.e., in block **1008**), (ii) the medium intermediate section mold unit (i.e., in block **1010**), or (iii) the large intermediate section mold unit **836** (i.e., in block **1012**). Selection of one of the intermediate mold units **832**, **834**, **836** as the first mold unit is described in greater detail below with reference to FIG. **11**. In any case, from block **1004**, the method **1000** subsequently proceeds to block **1014**.

(58) In block **1014** of the illustrative method **1000**, the operator(s) or the control system couples the selected first mold unit to the front cage mold unit **810** of the modular system **800**. It should be appreciated that to perform block **1014**, the selected first mold unit (i.e., the rear floor mold unit **820**) is coupled to the front cage mold unit **810** such that the front cage mold cavity **912** is fluidly coupled to the rear floor mold cavity **922** to at least partially establish a continuous monocoque mold cavity. Following performance of block **1014**, the method **1000** proceeds to block **1016**.

(59) In block **1016** of the illustrative method **1000**, the operator(s) or the control system introduces one or more composite materials (e.g., the composite materials included in the composite structure **700**) into the continuous monocoque mold cavity formed in block **1014**. More specifically, to perform block **1016**, at least in some embodiments, the operator(s) or the control system performs blocks **1018**, **1020**, and **1022**. In block **1018**, the operator(s) or the control system introduces one or more composite materials into the continuous monocoque mold cavity without introducing metallic material into the cavity. In other embodiments, however, block **1018** may be omitted from the method **1000**. In block **1020**, the operator(s) or the control system places a first material in the continuous monocoque mold cavity. The first material may include balsa wood and/or plastic, at least in some embodiments. In block **1022**, the operator(s) or the control system places a second material different from the first material in the continuous monocoque mold cavity. The second material may include fiberglass and resin, at least in some embodiments. Following performance of block **1016**, the method **1000** proceeds to block **1024**.

(60) In block **1024** of the illustrative method **1000**, the operator(s) or the control system cures the one or more composite materials in the continuous monocoque mold cavity to form the monocoque. To perform block **1024**, the operator(s) or the control system may perform blocks **1026**, **1028**, and **1030**, at least in some embodiments. In block **1026**, the operator(s) or the control system forms a core (e.g., the core **702**) including the first material introduced in block **1016**. In block **1028**, the operator(s) or the control system forms a shell (e.g., the shell **704**) including the second material introduced in block **1016** that at least partially surrounds the core. In block **1030**, the operator(s) or the control system forms a laminate layer (e.g., the layer **706**) that at least partially covers the shell.

(61) Referring now to FIGS. **11** and **12**, an illustrative method **1100** of forming a monocoque (e.g., the monocoque **200**) using a modular mold system (e.g., the system **800**) is depicted. The method **1100** corresponds to, or is otherwise associated with, performance of the blocks described below in the illustrative sequence of FIGS. **11** and **12**. It should be appreciated, however, that the method **1100** may be performed in one or more sequences different from the illustrative sequence. Furthermore, it should be appreciated that one or more of the blocks described below may be executed contemporaneously and/or in parallel with one another. In some embodiments, the method **1100** may be performed manually by one or more operators. In other embodiments, the method **1100** may be embodied as, or otherwise include, a set of instructions that are performed by an automated control system.

(62) The illustrative method **1100** begins with block **1102**. In block **1102**, the operator(s) or the control system selects a land vehicle type or a monocoque configuration for a particular land vehicle. It should be appreciated that to perform block **1102**, the operator(s) or the control system may select any vehicle envisioned by the present disclosure or any monocoque configuration associated with a particular vehicle contemplated by the present disclosure. From block **1102**, the method **1100** subsequently proceeds to block **1104**.

(63) In block **1104** of the illustrative method **1100**, the operator(s) or the control system selects a

first mold unit of the modular mold system based upon the selected vehicle type or monocoque configuration. In the illustrative embodiment, to perform block **1104**, the operator(s) or the control system performs one of blocks **1106**, **1108**, and **1110**. In block **1106**, the operator(s) or the control system selects the small intermediate section mold unit **832**. In block **1108**, the operator(s) or the control system selects the medium intermediate section mold unit **834**. In block **1110**, the operator(s) or the control system selects the large intermediate section mold unit **836**. Following performance of block **1104**, the method **1100** proceeds to block **1112**.

(64) In block **1112** of the illustrative method **1100**, the operator(s) or the control system selects a second mold unit of the modular system. In the illustrative embodiment, to perform block **1112**, the operator(s) or the control system performs block **1114**. In block **1114**, the operator(s) or the control system selects the rear floor mold unit **820** of the modular system **800**. From block **1112**, the method **1100** subsequently proceeds to block **1116**.

(65) In block **1116** of the illustrative method **1100**, the operator(s) or the control system couples the selected first mold unit to the front cage mold unit **810** of the modular system **800**. It should be appreciated that to perform block **1116**, the selected first mold unit (i.e., one of the intermediate mold units **832**, **834**, **836**) is coupled to the front cage mold unit **810** such that the front cage mold cavity **912** is fluidly coupled to the corresponding intermediate mold unit cavity (i.e., one of the cavities **933**, **935**, **937**) to at least partially establish a continuous monocoque mold cavity. Following performance of block **1116**, the method **1100** proceeds to block **1118**.

(66) In block **1118** of the illustrative method **1100**, the operator(s) or the control system couples the selected first mold unit (i.e., one of the intermediate mold units **832**, **834**, **836**) to the selected second mold unit (i.e., the rear floor mold unit **820**). It should be appreciated that to perform block **1118**, one of the intermediate mold units **832**, **834**, **836** is coupled to the rear floor mold unit **820** such that the rear floor mold cavity **922** is fluidly coupled to the corresponding intermediate mold unit cavity (i.e., one of the cavities **933**, **935**, **937**) to at least partially establish the continuous monocoque mold cavity. Following performance of block **1118**, the method **1100** proceeds to block **1120**.

(67) In block **1120** of the illustrative method **1100**, the operator(s) or the control system introduces one or more composite materials (e.g., the composite materials included in the composite structure **700**) into the continuous monocoque mold cavity formed in block **1118**. More specifically, to perform block **1120**, at least in some embodiments, the operator(s) or the control system performs blocks **1122**, **1124**, and **1126**. In block **1122**, the operator(s) or the control system introduces one or more composite materials into the continuous monocoque mold cavity without introducing metallic material into the cavity. In other embodiments, however, block **1122** may be omitted from the method **1100**. In block **1124**, the operator(s) or the control system places a first material in the continuous monocoque mold cavity. The first material may include balsa wood and/or plastic, at least in some embodiments. In block **1126**, the operator(s) or the control system places a second material different from the first material in the continuous monocoque mold cavity. The second material may include fiberglass and resin, at least in some embodiments. Following performance of block **1120**, the method **1000** proceeds to block **1202**.

(68) In block **1202** of the illustrative method **1100**, the operator(s) or the control system cures the one or more composite materials in the continuous monocoque mold cavity to form the monocoque. To perform block **1202**, the operator(s) or the control system may perform blocks **1204**, **1206** and **1208**, at least in some embodiments. In block **1204**, the operator(s) or the control system forms a core (e.g., the core **702**) including the first material introduced in block **1120**. In block **1206**, the operator(s) or the control system forms a shell (e.g., the shell **704**) including the second material introduced in block **1120** that at least partially surrounds the core. In block **1208**, the operator(s) or the control system forms a laminate layer (e.g., the layer **706**) that at least partially covers the shell.

(69) Referring now to FIG. **13**, an illustrative method **1300** of forming multiple monocoques of

land vehicles using at least one modular mold system is depicted. The method **1300** corresponds to, or is otherwise associated with, performance of the blocks described below in the illustrative sequence of FIG. **13**. It should be appreciated, however, that the method **1300** may be performed in one or more sequences different from the illustrative sequence. Furthermore, it should be appreciated that one or more of the blocks described below may be executed contemporaneously and/or in parallel with one another. In some embodiments, the method **1300** may be performed manually by one or more operators. In other embodiments, the method **1300** may be embodied as, or otherwise include, a set of instructions that are performed by an automated control system.

(70) The illustrative method **1300** begins with block **1302**. In block **1302**, the operator(s) or the control system forms a first monocoque of a first land vehicle. To perform block **1302**, the operator(s) or the control system forms the first monocoque of the first land vehicle using at least one modular system (e.g., the system **800**) in block **1304**. In some embodiments, the first monocoque of the first land vehicle is formed using only the front cage mold unit **810** and the rear floor mold unit **820** of the modular system **800**. In those embodiments, the first monocoque of the first land vehicle may be formed by performing the method **1000** described above. In other embodiments, the first monocoque of the first land vehicle is formed using the front cage mold unit **810**, the rear floor mold unit **820**, and one of the intermediate mold units **832**, **834**, **836**. In those embodiments, the first monocoque of the first land vehicle may be formed by performing the method **1100** described above. In any case, following performance of block **1302**, the method **1300** proceeds to block **1306**.

(71) In block **1306** of the illustrative method **1300**, the operator(s) or the control system forms a second monocoque of a second land vehicle that is different from the first land vehicle. To perform block **1306**, the operator(s) or the control system forms the second monocoque of the second land vehicle using at least one modular system (i.e., the system **800**) in block **1308**. In embodiments in which the first monocoque of the first land vehicle is formed in block **1302** using only the front cage mold unit **810** and the rear floor mold unit **820** of the modular system **800** (i.e., according to the method **1000**), the second monocoque of the second land vehicle is formed using the front cage mold unit **810**, the rear floor mold unit **820**, and one of the intermediate mold units **832**, **834**, **836** (i.e., according to the method **1100**). In embodiments in which the first monocoque of the first land vehicle is formed in block **1302** using the front cage mold unit **810**, the rear floor mold unit **820**, and a first one of the intermediate mold units **832**, **834**, **836** (i.e., according to the method **1100**), the second monocoque of the second land vehicle is formed using the front cage mold unit **810**, the rear floor mold unit **820**, and a second one of the intermediate mold units **832**, **834**, **836** that is different from the first one. Regardless, from block **1306**, the method **1300** subsequently proceeds to block **1310**.

(72) In block **1310** of the illustrative method **1300**, the operator(s) or the control system forms a third monocoque of a third land vehicle that is different from the first land vehicle and the second land vehicle. To perform block **1310**, the operator(s) or the control system forms the third monocoque of the third land vehicle using at least one modular system (i.e., the system **800**) in block **1310**. In embodiments in which (i) the first monocoque of the first land vehicle is formed in block **1302** using only the front cage mold unit **810** and the rear floor mold unit **820** of the modular system **800** (i.e., according to the method **1000**) and (ii) the second monocoque of the second land vehicle is formed in block **1306** using the front cage mold unit **810**, the rear floor mold unit **820**, and a first one of the intermediate mold units **832**, **834**, **836** (i.e., according to the method **1100**), the third monocoque of the third land vehicle is formed using the front cage mold unit **810**, the rear floor mold unit **820**, and a second one of the intermediate mold units **832**, **834**, **836** that is different from the first one. In embodiments in which (i) the first monocoque of the first land vehicle is formed in block **1302** using the front cage mold unit **810**, the rear floor mold unit **820**, and a first one of the intermediate mold units **832**, **834**, **836** (i.e., according to the method **1100**) and (ii) the second monocoque of the second land vehicle is formed in block **1306** using the front cage mold

unit **810**, the rear floor mold unit **820**, and a second one of the intermediate mold units **832**, **834**, **836** that is different from the first one, the third monocoque of the third land vehicle is formed using the front cage mold unit **810**, the rear floor mold unit **820**, and a third one of the intermediate mold units **832**, **834**, **836** that is different from the first one and the second one.

(73) Referring now to FIG. **14**, the six-passenger flatbed utility vehicle **140** shown in FIG. **1** is depicted in greater detail. It should be appreciated that the illustrative vehicle **140** is an electric vehicle as mentioned above. In the illustrative embodiment, the vehicle **140** includes a front cage **1410**, a rear floor **1430**, an intermediate section **1450**, and a flatbed **1470**. The front cage **1410** at least partially defines an operator cabin **1412** of the vehicle **140**. The rear floor **1430** is positioned rearward of the front cage **1410** in a longitudinal direction **D1**. The intermediate section **1450** is disposed at least partially between the front cage **1410** and the rear floor **1430** in the longitudinal direction **D1**. The flatbed **1470** is at least partially defined by the rear floor **1430** and open to the ambient environment. The flatbed **1470** defines a utility space **1472** that is located outside the operator cabin **1412** and accessible from a rear end **1404** of the vehicle **140**.

(74) The illustrative vehicle **140** may be formed from the monocoque system **900** and/or the modular mold system **800** described herein. The front cage **1410** may be formed from the front cage mold unit **810**, the rear floor **1430** and the flatbed **1470** may be formed from the rear floor mold unit **820**, and the intermediate section **1450** may be formed from one of the intermediate mold units **832**, **834**, **836**, at least in some embodiments. Additionally, in some embodiments, the front cage **1410** may be formed from the front cage mold unit **810** having the shape and structure of the mold unit **910**, the rear floor **1430** and the flatbed **1470** may be formed from the rear floor mold unit **820** having the shape and structure of the mold unit **920**, and the intermediate section **1450** may be formed from one of the intermediate mold units **832**, **834**, **836** having the shape and structure of the corresponding units **932**, **934**, **936**.

(75) In the illustrative embodiment, the vehicle **140** includes a utility cabinet **1460** having drawers **1462** that are accessible from the utility space **1472**. The utility cabinet **1460** is arranged at least partially between the front cage **1410** and the rear floor **1430** in the longitudinal direction **D1**. As such, in at least some embodiments, the utility cabinet **1460** is aligned with the intermediate section **1450** in the longitudinal direction **D1**. Additionally, in some embodiments, the utility cabinet **1460** is included in, or otherwise forms a portion of, the intermediate section **1450**.

(76) The illustrative utility cabinet **1460** extends forward (i.e., toward a front end **1402** of the vehicle **140**) in the longitudinal direction **D1** to the front cage **1410**. In some embodiments, the utility cabinet **1460** extends at least partially into the operator cabin **1412**. In such embodiments, the operator cabin **1412** may define an interior space **1414** (shown in phantom) that receives a portion of the utility cabinet **1460**, and positioning of the utility cabinet **1460** in the interior space **1414** may close off, or substantially close off, the interior space **1414** from fluid communication with the utility space **1472**.

(77) In some embodiments, the utility cabinet **1460** extends forward in the longitudinal direction **D1** to the front cage **1410** such that the utility cabinet **1460** is flush with, and/or in contact with, a wall **1416** of the front cage **1410** that closes off, or substantially closes off, the operator cabin **1412** from fluid communication with the utility space **1472**. In such embodiments, the utility cabinet **1460** may not extend into the operator cabin **1412**. In such embodiments, the utility cabinet **1460** may be disposed entirely within the utility space **1472**.

(78) Each of the drawers **1462** of the utility cabinet **1460** are depicted in FIG. **14** in a closed state **1464**. When the drawers **1462** are in the closed state **1464**, the drawers **1462** are aligned with the intermediate section **1450** in the longitudinal direction **D1**. As suggested by FIG. **14**, when each of the drawers **1462** is in an opened state **1466** (shown in phantom), each of the drawers **1462** extends rearward (i.e., toward the rear end **1404**) in the longitudinal direction **D1** into the utility space **1472**. Therefore, in at least some embodiments, the drawers **1462** may be at least partially aligned with the rear floor **1430** and the flatbed **1470** in the longitudinal direction **D1** in the opened states **1466**.

thereof.

(79) In some embodiments, the utility cabinet **1460** directly contacts the wall **1416** as indicated above. The wall **1416** illustratively includes, or is otherwise embodied as, a rearwardly-facing exterior wall of the front cage **1410**. The wall **1416** extends in a vertical direction VD to a roof **1418** of the front cage **1410** such that the wall **1416** is interconnected with the roof **1418**. The wall **1416** extends above a top or uppermost surface **1468** of the utility cabinet **1460** to the roof **1418**. The wall **1416**, the roof **1418**, and the top surface **1468** cooperate to define a storage space **1474** that is accessible from, and in fluid communication with, the utility space **1472**.

(80) The illustrative flatbed **1470** of the vehicle **140** includes sidewalls **1476**, **1478** arranged opposite one another, a rear gate **1480**, a rear bumper **1482**, and a vice **1484**. Each of the sidewalls **1476**, **1478** extends upwardly away from the rear floor **1430** in the vertical direction VD to at least partially define the utility space **1472**. Each of the sidewalls **1476**, **1478** is formed without an opening extending therethrough. The rear gate **1480** extends between the sidewalls **1476**, **1478** in a lateral direction D2 perpendicular to the longitudinal direction D1 to at least partially close off the utility space **1472**. The rear bumper **1482** extends rearwardly in the longitudinal direction D1 away from the rear gate **1480**. The vice **1484** is mounted to the rear bumper **1482**.

(81) In the illustrative embodiment, the flatbed **1470** includes a guide rail **1486** interconnected with the sidewall **1476**. The guide rail **1486** extends upwardly away from the sidewall **1476** in the vertical direction VD to define a slot **1488** between the sidewall **1476** and the guide rail **1486**. In some embodiments, the guide rail **1486** provides a handle that may be grasped by a subject to provide stability and/or support when the subject is positioned in the utility space **1472** or when the subject is positioned in close proximity to the guide rail **1486** outside the utility space **1472**. Additionally, in some embodiments, the guide rail **1486** provides a structure to which other devices (e.g., utility devices, accessories, or articles disposed in the utility space **1472**) may be coupled and/or secured to facilitate transport of those devices in use of the vehicle **140**.

(82) In the illustrative embodiment, the flatbed **1470** includes a guide rail **1490** interconnected with the sidewall **1478**. The guide rail **1490** extends upwardly away from the sidewall **1478** in the vertical direction VD to define a slot **1492** between the sidewall **1478** and the guide rail **1490**. In some embodiments, the guide rail **1490** provides a handle that may be grasped by a subject to provide stability and/or support when the subject is positioned in the utility space **1472** or when the subject is positioned in close proximity to the guide rail **1490** outside the utility space **1472**. Additionally, in some embodiments, the guide rail **1490** provides a structure to which other devices (e.g., utility devices, accessories, or articles disposed in the utility space **1472**) may be coupled and/or secured to facilitate transport of those devices in use of the vehicle **140**.

(83) It should be appreciated that a variety of devices may be arranged and/or transported in the utility space **1472** of the vehicle **140** in use thereof. In the illustrative example of FIG. **14**, a cable spool **1494**, a tub **1496**, and a cooler **1498** are disposed in the utility space **1472**. The cable spool **1494**, the tub **1496**, and the cooler **1498** may be tied or tethered to one of the guide rails **1486**, **1490**, at least in some embodiments. Of course, in other embodiments, other devices may be positioned in the utility space **1472** and secured to one of the guide rails **1486**, **1490** depending on the particular mission or application of the vehicle **140**.

(84) In some embodiments, the flatbed **1470** includes a bench **1495** arranged in the utility space **1472** and supported by the rear floor **1430**. The illustrative bench **1495** is disposed beneath the utility cabinet **1460** in the vertical direction VD. As a result, the drawers **1462** of the utility cabinet **1460** may be opened and closed without interference with the bench **1495**. In some embodiments, the bench **1495** may be aligned in the vertical direction VD with a base (not shown) of the utility cabinet **1460** located beneath the drawers **1462**.

(85) The illustrative bench **1495** includes a leg **1497** arranged to extend laterally through the utility space **1472** in the lateral direction D2. The bench **1495** also includes a leg **1499** interconnected with the leg **1497** that is arranged to extend longitudinally through the utility space **1472** in the

longitudinal direction D1. In some embodiments, the leg **1497** extends laterally through the utility space **1472** over an entire width W of the utility space **1472**. In other embodiments, the leg **1497** extends laterally through the utility space **1472** over part of the width W of the utility space **1472**. In some embodiments, the leg **1499** extends longitudinally through the utility space **1472** over an entire length L of the utility space **1472**. In other embodiments, the leg **1499** extends longitudinally through the utility space **1472** over part of the length L of the utility space **1472**.

(86) In the illustrative embodiment, the intermediate section **1450** has a fixed, single length (i.e., in the longitudinal direction D1) and is not adjustable to vary the length thereof. The intermediate section **1450** includes at least one reinforcement beam **1452** that extends between the roof **1418** and the flatbed **1470**. More specifically, the illustrative reinforcement beam **1452** is interconnected with the roof **1418** at one end **1454** thereof and interconnected with the sidewall **1476** at another end **1456** thereof opposite the end **1454**. In some embodiments, the intermediate section **1450** may include another reinforcement beam (not shown) that is interconnected with the roof **1418** at one end thereof and interconnected with the sidewall **1478** at another end thereof opposite the one end.

(87) The illustrative reinforcement beam **1452** extends oblique to the sidewall **1476** to define an obtuse angle α between the reinforcement beam **1452** and the sidewall **1476**. In the illustrative embodiment, the reinforcement beam **1452** is directly interconnected with the guide rail **1486**. More specifically, the reinforcement beam **1452** is directly interconnected with the guide rail **1486** in close proximity to an alcove **1458** of the intermediate section **1450**.

(88) The reinforcement beam **1452** at least partially defines the alcove **1458** of the intermediate section **1450**. The alcove **1458** is disposed at least partially between the front cage **1410** and the rear floor **1430** in the longitudinal direction D1. The alcove **1458** is illustratively aligned with the utility cabinet **1460** in the longitudinal direction D1. In the illustrative embodiment, the alcove **1458** defines an opening **1459** into the intermediate section **1450** in the lateral direction D2. The opening **1459** is closed off from the operator cabin **1412**. In some embodiments, the alcove **1458** includes, or is otherwise embodied as, a nook accessible from the utility space **1472** or from outside the vehicle **140** in close proximity to the intermediate section **1450**. Additionally, in some embodiments, the alcove **1458** provides an exterior compartment for storing items, such as a fire extinguisher, for example.

(89) In the illustrative embodiment, the front cage **1410** includes doors **1420**, **1422** arranged on a driver side **1424** of the vehicle **140**. It should be appreciated that corresponding doors (not shown) are arranged on a passenger side **1426** of the vehicle **140** disposed opposite the driver side **1424**. The doors **1420**, **1422** are illustratively arranged between the front end **1402** of the vehicle **140** and the alcove **1458** in the longitudinal direction D1. The door **1420** is positioned forward of the door **1422** in the longitudinal direction D1. The doors **1420**, **1422** and the doors disposed on the passenger side **1426** cooperate to at least partially define the operator cabin **1412**, which is configured to accommodate six passengers as indicated above.

(90) The illustrative vehicle **140** includes a monocoque **1406** that at least partially defines the front cage **1410**, the rear floor **1430**, the intermediate section **1450**, and the flatbed **1470**. The monocoque **1406** illustratively includes, or is otherwise embodied as, a single-piece, monolithic structure unsupported by an internal chassis. Put differently, the monocoque **1406** does not include an internal chassis, at least in some embodiments. The monocoque **1406** has a composite construction such that the front cage **1410**, the rear floor **1430**, the intermediate section **1450**, and the flatbed **1470** are formed from one or more composite materials. In some embodiments, the monocoque **1406** has the composite structure **700** described above. Additionally, in some embodiments, the monocoque **1406** does not include metallic material.

(91) Referring now to FIG. **15**, the two-passenger flatbed utility vehicle **110** shown in FIG. **1** is depicted in greater detail. It should be appreciated that the illustrative vehicle **110** is an electric vehicle as mentioned above. In the illustrative embodiment, the vehicle **110** includes a front cage **1510**, a rear floor **1530**, an intermediate section **1550**, and a flatbed **1570**. The front cage **1510** at

least partially defines an operator cabin **1512** of the vehicle **110**. The rear floor **1530** is positioned rearward of the front cage **1510** in a longitudinal direction **D1'**. The intermediate section **1550** is disposed at least partially between the front cage **1510** and the rear floor **1530** in the longitudinal direction **D1'**. The flatbed **1570** is at least partially defined by the rear floor **1530** and open to the ambient environment. The flatbed **1570** defines a utility space **1572** that is located outside the operator cabin **1512** and accessible from a rear end **1504** of the vehicle **110**.

(92) The illustrative vehicle **110** may be formed from the monocoque system **900** and/or the modular mold system **800** described herein. The front cage **1510** may be formed from the front cage mold unit **810**, the rear floor **1530** and the flatbed **1570** may be formed from the rear floor mold unit **820**, and the intermediate section **1550** may be formed from one of the intermediate mold units **832**, **834**, **836**, at least in some embodiments. Additionally, in some embodiments, the front cage **1510** may be formed from the front cage mold unit **810** having the shape and structure of the mold unit **910**, the rear floor **1530** and the flatbed **1570** may be formed from the rear floor mold unit **820** having the shape and structure of the mold unit **920**, and the intermediate section **1550** may be formed from one of the intermediate mold units **832**, **834**, **836** having the shape and structure of the corresponding units **932**, **934**, **936**.

(93) The illustrative flatbed **1570** of the vehicle **140** includes sidewalls **1576**, **1578** arranged opposite one another, a rear gate **1580**, and a rear bumper **1582**. Each of the sidewalls **1576**, **1578** extends upwardly away from the rear floor **1530** in the vertical direction **VD'** to at least partially define the utility space **1572**. Each of the sidewalls **1576**, **1578** is formed without an opening extending therethrough. The rear gate **1580** extends between the sidewalls **1576**, **1578** in a lateral direction **D2'** perpendicular to the longitudinal direction **D1'** to at least partially close off the utility space **1572**. The rear bumper **1582** extends rearwardly in the longitudinal direction **D1'** away from the rear gate **1580**.

(94) In the illustrative embodiment, the flatbed **1570** includes a guide rail **1586** interconnected with the sidewall **1576**. The guide rail **1586** extends upwardly away from the sidewall **1576** in the vertical direction **VD'** to define a slot **1588** between the sidewall **1576** and the guide rail **1586**. In some embodiments, the guide rail **1586** provides a handle that may be grasped by a subject to provide stability and/or support when the subject is positioned in the utility space **1572** or when the subject is positioned in close proximity to the guide rail **1586** outside the utility space **1572**. Additionally, in some embodiments, the guide rail **1586** provides a structure to which other devices (e.g., utility devices, accessories, or articles disposed in the utility space **1572**) may be coupled and/or secured to facilitate transport of those devices in use of the vehicle **110**.

(95) In the illustrative embodiment, the flatbed **1570** includes a guide rail **1590** interconnected with the sidewall **1578**. The guide rail **1590** extends upwardly away from the sidewall **1578** in the vertical direction **VD'** to define a slot **1592** between the sidewall **1578** and the guide rail **1590**. In some embodiments, the guide rail **1590** provides a handle that may be grasped by a subject to provide stability and/or support when the subject is positioned in the utility space **1572** or when the subject is positioned in close proximity to the guide rail **1590** outside the utility space **1572**. Additionally, in some embodiments, the guide rail **1590** provides a structure to which other devices (e.g., utility devices, accessories, or articles disposed in the utility space **1572**) may be coupled and/or secured to facilitate transport of those devices in use of the vehicle **110**.

(96) It should be appreciated that a variety of devices may be arranged and/or transported in the utility space **1572** of the vehicle **110** in use thereof. Such devices may include the devices **1494**, **1496**, **1498** described above with reference to FIG. **14**, which may be tied or tethered to one of the guide rails **1586**, **1590**, at least in some embodiments. Of course, in other embodiments, other devices may be positioned in the utility space **1572** and secured to one of the guide rails **1586**, **1590** depending on the particular mission or application of the vehicle **110**.

(97) In the illustrative embodiment, the intermediate section **1550** has a fixed, single length (i.e., in the longitudinal direction **D1'**) and is not adjustable to vary the length thereof. The intermediate

section **1550** includes at least one reinforcement beam **1552** that extends between the roof **1518** and the flatbed **1570**. More specifically, the illustrative reinforcement beam **1552** is interconnected with the roof **1518** at one end **1554** thereof and interconnected with the sidewall **1576** at another end **1556** thereof opposite the end **1554**. In some embodiments, the intermediate section **1550** may include another reinforcement beam (not shown) that is interconnected with the roof **1518** at one end thereof and interconnected with the sidewall **1578** at another end thereof opposite the one end.

(98) The illustrative reinforcement beam **1552** extends oblique to the sidewall **1576** to define an obtuse angle B between the reinforcement beam **1552** and the sidewall **1576**. In the illustrative embodiment, the reinforcement beam **1552** is directly interconnected with the guide rail **1586**. More specifically, the reinforcement beam **1552** is directly interconnected with the guide rail **1586** in close proximity to an alcove **1558** of the intermediate section **1550**.

(99) The reinforcement beam **1552** at least partially defines the alcove **1558** of the intermediate section **1550**. The alcove **1558** is disposed at least partially between the front cage **1510** and the rear floor **1530** in the longitudinal direction D1'. In the illustrative embodiment, the alcove **1558** defines an opening **1559** into the intermediate section **1550** in the lateral direction D2'. The opening **1559** is closed off from the operator cabin **1512**. In some embodiments, the alcove **1558** includes, or is otherwise embodied as, a nook accessible from the utility space **1572** or from outside the vehicle **110** in close proximity to the intermediate section **1550**. Additionally, in some embodiments, the alcove **1558** provides an exterior compartment for storing items, such as a fire extinguisher, for example.

(100) In the illustrative embodiment, the front cage **1510** includes a door **1520** arranged on a driver side **1524** of the vehicle **110**. It should be appreciated that a corresponding door (not shown) is arranged on a passenger side **1526** of the vehicle **110** disposed opposite the driver side **1524**. The door **1520** is illustratively arranged between the front end **1502** of the vehicle **110** and the alcove **1558** in the longitudinal direction D1'. The door **1520** and the door disposed on the passenger side **1526** cooperate to at least partially define the operator cabin **1512**, which is configured to accommodate two passengers as indicated above.

(101) The illustrative vehicle **110** includes a monocoque **1506** that at least partially defines the front cage **1510**, the rear floor **1530**, the intermediate section **1550**, and the flatbed **1570**. The monocoque **1506** illustratively includes, or is otherwise embodied as, a single-piece, monolithic structure unsupported by an internal chassis. Put differently, the monocoque **1506** does not include an internal chassis, at least in some embodiments. The monocoque **1506** has a composite construction such that the front cage **1510**, the rear floor **1530**, the intermediate section **1550**, and the flatbed **1570** are formed from one or more composite materials. In some embodiments, the monocoque **1506** has the composite structure **700** described above. Additionally, in some embodiments, the monocoque **1506** does not include metallic material.

(102) While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

Claims

1. An electric vehicle comprising: one or more electric motors; a front cage at least partially defining an operator cabin that includes a roof and a rear wall; and a flatbed positioned rearward of the front cage in a longitudinal direction that comprises a first sidewall interconnected with the roof by a first reinforcement beam, wherein: the first reinforcement beam and the rear wall cooperate to at least partially define a cabinet storage space, and a utility cabinet is arranged in the cabinet storage space.
2. The electric vehicle of claim 1, wherein the rear wall closes off the operator cabin from a utility

space defined by the flatbed.

3. The electric vehicle of claim 1, wherein the utility cabinet is arranged in the cabinet storage space such that the utility cabinet is flush with the rear wall.
4. The electric vehicle of claim 1, wherein the first reinforcement beam extends oblique to the first sidewall to define an obtuse angle between the first reinforcement beam and the first sidewall.
5. The electric vehicle of claim 1, further comprising an alcove that extends in a lateral direction through an exterior wall of the vehicle to define an exterior compartment accessible from outside the vehicle, and wherein the exterior compartment opens into the cabinet storage space.
6. The electric vehicle of claim 5, wherein the first reinforcement beam at least partially defines the alcove.
7. The electric vehicle of claim 5, wherein the alcove and the cabinet storage space are aligned in the longitudinal direction.
8. The electric vehicle of claim 5, wherein the exterior compartment is closed off from the operator cabin.
9. The electric vehicle of claim 1, wherein the flatbed includes a second sidewall arranged opposite the first sidewall that is interconnected with the roof by a second reinforcement beam arranged opposite the first reinforcement beam.
10. The electric vehicle of claim 9, wherein the cabinet storage space is at least partially defined by the rear wall, the first reinforcement beam, and the second reinforcement beam.
11. An electric vehicle comprising: one or more electric motors; a front cage at least partially defining an operator cabin that includes a rear wall; and a flatbed positioned rearward of the front cage in a longitudinal direction that comprises a first sidewall coupled to the rear wall by a first reinforcement beam, wherein: the first reinforcement beam and the rear wall cooperate to at least partially define a cabinet storage space, a utility cabinet is arranged in the cabinet storage space, and the first reinforcement beam extends oblique to the first sidewall to define an obtuse angle between the first reinforcement beam and the first sidewall.
12. The electric vehicle of claim 11, wherein the utility cabinet is arranged in the cabinet storage space such that the utility cabinet is flush with the rear wall and in close proximity to the first reinforcement beam.
13. The electric vehicle of claim 11, further comprising an alcove that extends in a lateral direction through an exterior wall of the vehicle to define an exterior compartment accessible from outside the vehicle, and wherein the exterior compartment opens into the cabinet storage space.
14. The electric vehicle of claim 13, wherein the first reinforcement beam at least partially defines the alcove.
15. The electric vehicle of claim 13, wherein the alcove and the cabinet storage space are aligned in the longitudinal direction.
16. The electric vehicle of claim 13, wherein the exterior compartment is closed off from the operator cabin.
17. The electric vehicle of claim 11, wherein the flatbed includes a second sidewall arranged opposite the first sidewall that is coupled to the rear wall by a second reinforcement beam arranged opposite the first reinforcement beam.
18. The electric vehicle of claim 17, wherein the cabinet storage space is at least partially defined by the rear wall, the first reinforcement beam, and the second reinforcement beam.
19. The electric vehicle of claim 17, wherein the second reinforcement beam extends oblique to the second sidewall to define an obtuse angle between the second reinforcement beam and the second sidewall.
20. An electric vehicle comprising: one or more electric motors; a front cage at least partially defining an operator cabin that includes a rear wall; and a flatbed positioned rearward of the front cage in a longitudinal direction that comprises a first sidewall coupled to the rear wall by a first reinforcement beam and a second sidewall coupled to the rear wall by a second reinforcement

beam, wherein: the first reinforcement beam, the second reinforcement beam, and the rear wall cooperate to at least partially define a cabinet storage space, and a utility cabinet is arranged in the cabinet storage space.
